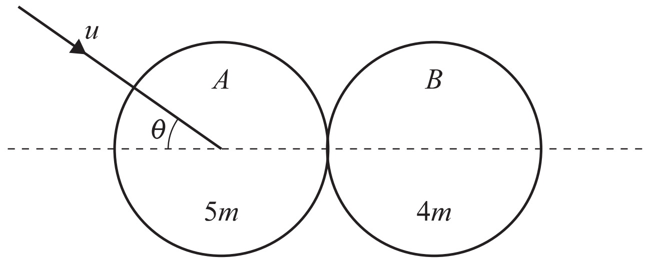
6



Two uniform smooth spheres A and B of equal radii have masses $5m$ and $4m$ respectively. Sphere A is moving with speed u on a horizontal surface when it collides with sphere B which is at rest. Immediately before the collision, A's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is e.

(a) Show that the speed of B after the collision is $\frac{5}{9}u(1+e)\cos\theta$. [3]

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After the collision the kinetic energy of A is equal to the kinetic energy of B.

(b) Given that $\tan\theta = \frac{2}{3}$, find the value of e. [6]